

Bloom Filter-Based MPSI

Weekly Progress Meeting 30 April 2026

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30-04-2026



Bibliography

- [1] Jelle Vos, Jorrit van Assen, Tjitske Koster, Evangelia Anna Markatou, and Zekeriya Erkin. On the insecurity of bloom filter-based private set intersections. *Cryptology ePrint Archive*, 2024.
- [2] Ou Ruan, Changwang Yan, Jing Zhou, and Chaohao Ai. A practical multiparty private set intersection protocol based on bloom filters for unbalanced scenarios. *Applied Sciences*, 13(24):13215, 2023.
- [3] Florian Kerschbaum. Outsourced private set intersection using homomorphic encryption. In *Proceedings of the 7th ACM Symposium on Information, Computer and Communications Security*, pages 85–86, 2012.
- [4] Falzon, Francesca, and Evangelia Anna Markatou. "Re-visiting authorized private set intersection: A new privacy-preserving variant and two protocols." *Proceedings on Privacy Enhancing Technologies* (2025).
- [5] Koster, Tjitske Ollie, Francesca Falzon, and Evangelia Anna Markatou. "Bandwidth Efficient Partial Authorized PSI." *Cryptology ePrint Archive* (2025).
- [6] Lindell, Yehuda. "How to simulate it—a tutorial on the simulation proof technique." *Tutorials on the Foundations of Cryptography: Dedicated to Oded Goldreich* (2017): 277-346.

Progress

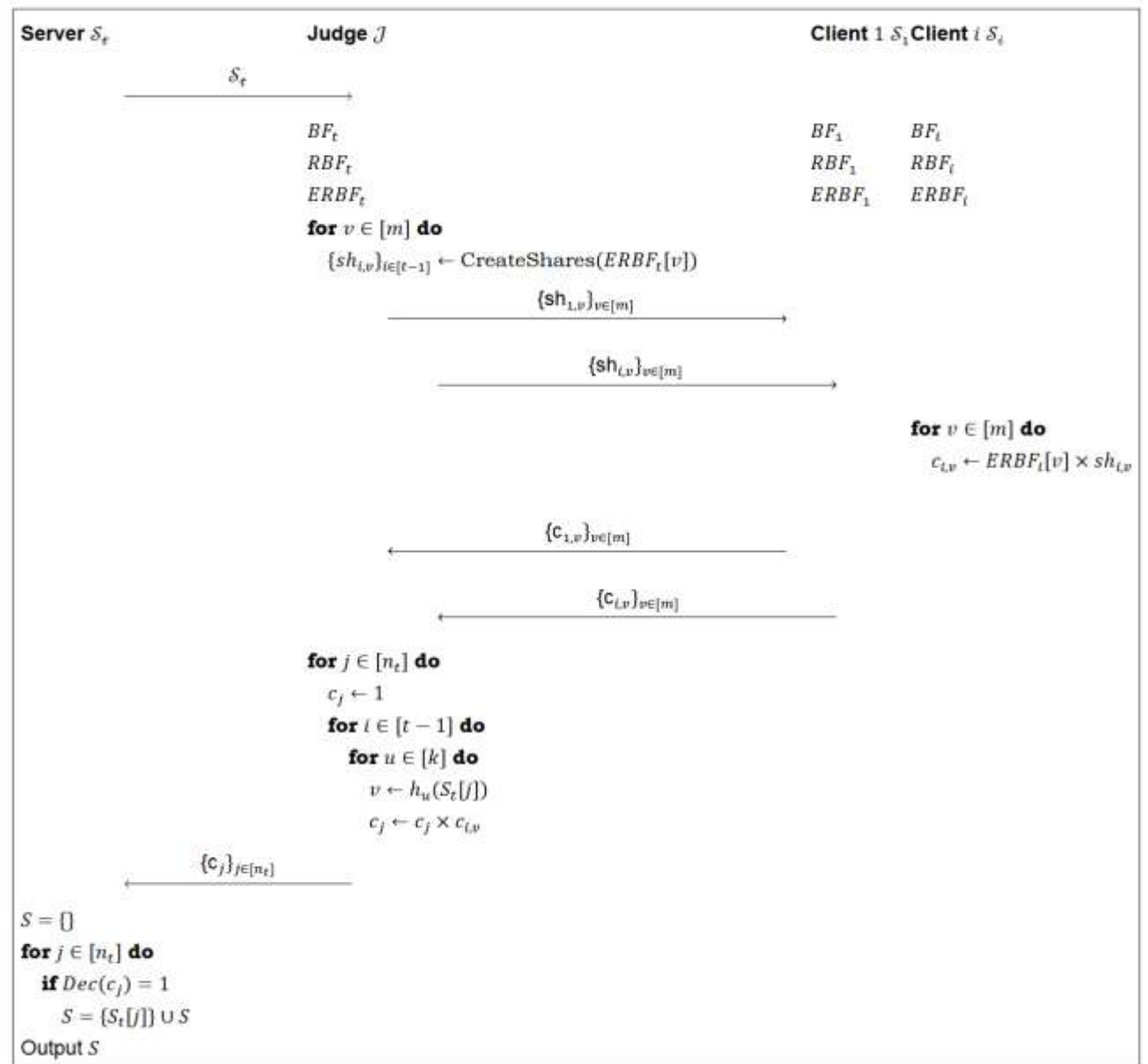
- Continue implementing APSI2+PAPSI2 / APSI3 / APSI4+PAPSI4 / **APSI5+PAPSI5**
- Writing!

Progress

APSI 2 – Active Judge *before*

Threat Model:

- Malicious Server
- Semi-honest Judge
- Semi-honest Clients

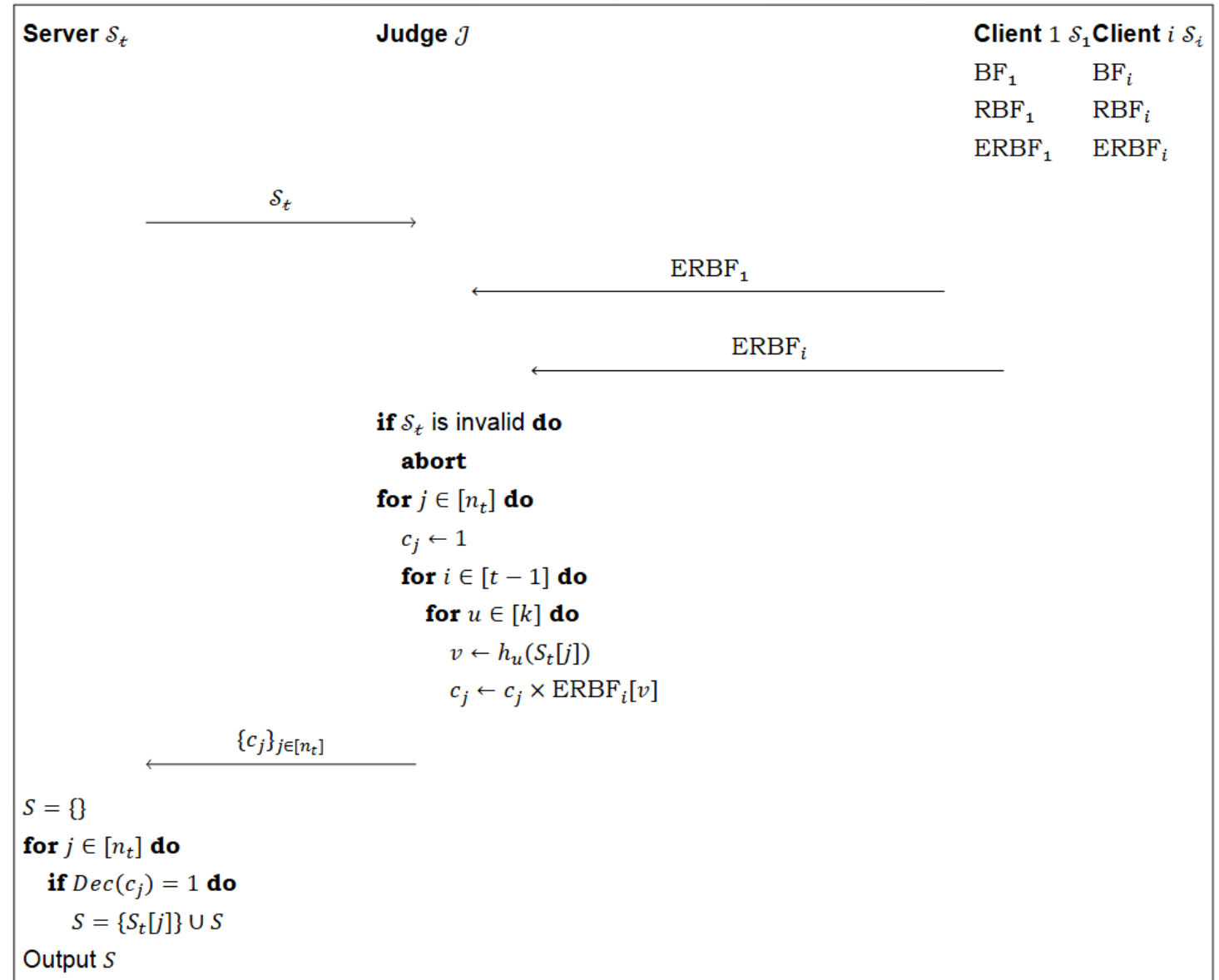


Progress

APSI 2 – Active Judge *after*

Threat Model:

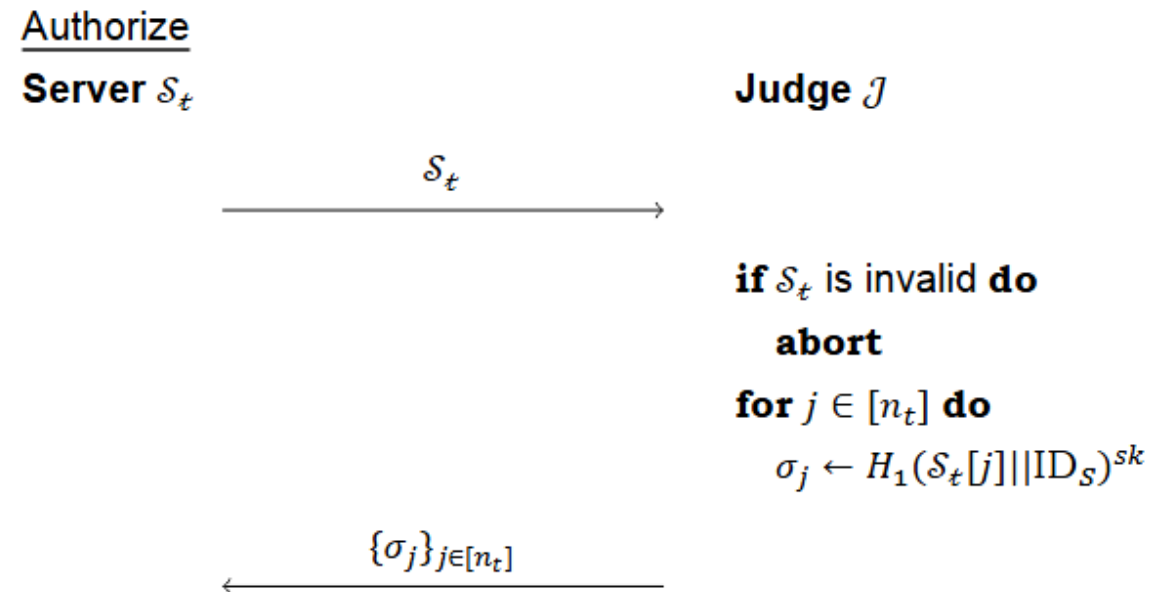
- Malicious Server
- Semi-honest Judge
- Semi-honest Clients



Progress

APSI 5 – Lilika's Garbled Judge

- Authorization is identical to [4], except the server will authorize their set since they are the one receiving the intersection



Progress

APSI 5 – Lilika's Garbled Judge

- We compute the GBFs by hashing the elements and secret sharing the bilinear pairing into the bins
- In GBFs, false positives turn into false negatives, since all their bins are occupied and we cannot secret share two different values using the same bins
 - Hence, the FP rate should be negligible (2^{-30})

Intersect
Server S_e

Client 1 S_1

...

Client i S_i

$s_1 \leftarrow \mathbb{Z}_q$

$S_1 \leftarrow g_2^{s_1}$

$\text{GBF}_1 \leftarrow \text{InitializeGBF}(m, k)$

for $j \in [n_1]$ **do**

$\hat{y}_{1,j} \leftarrow e(H_1(S_1[j] || \text{ID}_S), pk^{s_1})$

$\text{AddToGBF}(\text{GBF}_1, S_1[j], \hat{y}_{1,j})$

$s_i \leftarrow \mathbb{Z}_q$

$S_i \leftarrow g_2^{s_i}$

$\text{GBF}_i \leftarrow \text{InitializeGBF}(m, k)$

for $j \in [n_i]$ **do**

$\hat{y}_{i,j} \leftarrow e(H_1(S_i[j] || \text{ID}_S), pk^{s_i})$

$\text{AddToGBF}(\text{GBF}_i, S_i[j], \hat{y}_{i,j})$

S_2, GBF_2

S_i, GBF_i

$S_{\text{agg}} \leftarrow \prod_{i=1}^{t-1} S_i$

for $v \in [m]$ **do**

$\text{GBF}_{\text{agg}}[v] \leftarrow \prod_{i=1}^{t-1} \text{GBF}_i[v]$

$S_{\text{agg}}, \text{GBF}_{\text{agg}}$

$I = \{\}$

for $j \in [n_t]$ **do**

$\hat{x}_j \leftarrow e(\sigma_j, S_{\text{agg}})$

$\hat{y}_j \leftarrow \prod_{u=1}^k \text{GBF}_{\text{agg}}[h_u(S_t[j])]$

if $\hat{x}_j = \hat{y}_j$ **do**

$I = \{S_t[j]\} \cup I$

Output I

Progress

APSI 5 – Lilika's Garbled Judge

Proof outline against a semi-honest client leader:

- **H0:** Real protocol
- **H1:** Replace target values $\widehat{y}_{i,j}$ with random elements from $\mathbb{G}_T$
 - The leader will have to distinguish between $e(H_1(\mathcal{S}_i[j]||ID_S), g_2^{sk \times s_i})$ and a random element from $\mathbb{G}_T$, having $\mathcal{S}_i = g_2^{s_i}$ and $pk = g_2^{sk}$ - Decisional Bilinear Diffie Hellman (DBDH)
- **H2:** Replace $\mathcal{S}_i$ + all bins in the GBFs with random elements from $\mathbb{G}_T$

Intersect
Server $\mathcal{S}_e$

Client 1 $\mathcal{S}_1$

$s_1 \leftarrow \mathbb{Z}_q$
 $\mathcal{S}_1 \leftarrow g_2^{s_1}$
 $GBF_1 \leftarrow \text{InitializeGBF}(m, k)$
for $j \in [n_1]$ **do**
 $\hat{y}_{1,j} \leftarrow e(H_1(\mathcal{S}_1[j]||ID_S), pk^{s_1})$
 AddToGBF($GBF_1, \mathcal{S}_1[j], \hat{y}_{1,j}$)

...

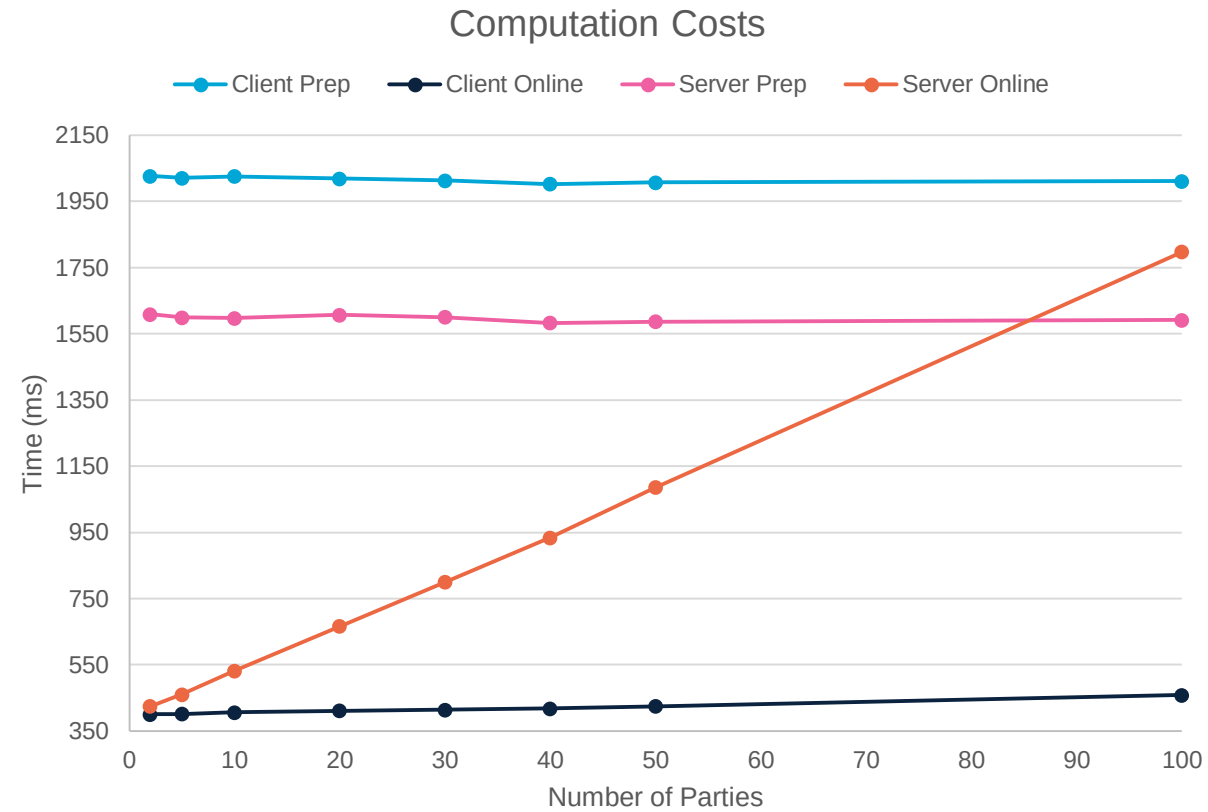
Client i $\mathcal{S}_i$

$s_i \leftarrow \mathbb{Z}_q$
 $\mathcal{S}_i \leftarrow g_2^{s_i}$
 $GBF_i \leftarrow \text{InitializeGBF}(m, k)$
for $j \in [n_i]$ **do**
 $\hat{y}_{i,j} \leftarrow e(H_1(\mathcal{S}_i[j]||ID_S), pk^{s_i})$
 AddToGBF($GBF_i, \mathcal{S}_i[j], \hat{y}_{i,j}$)

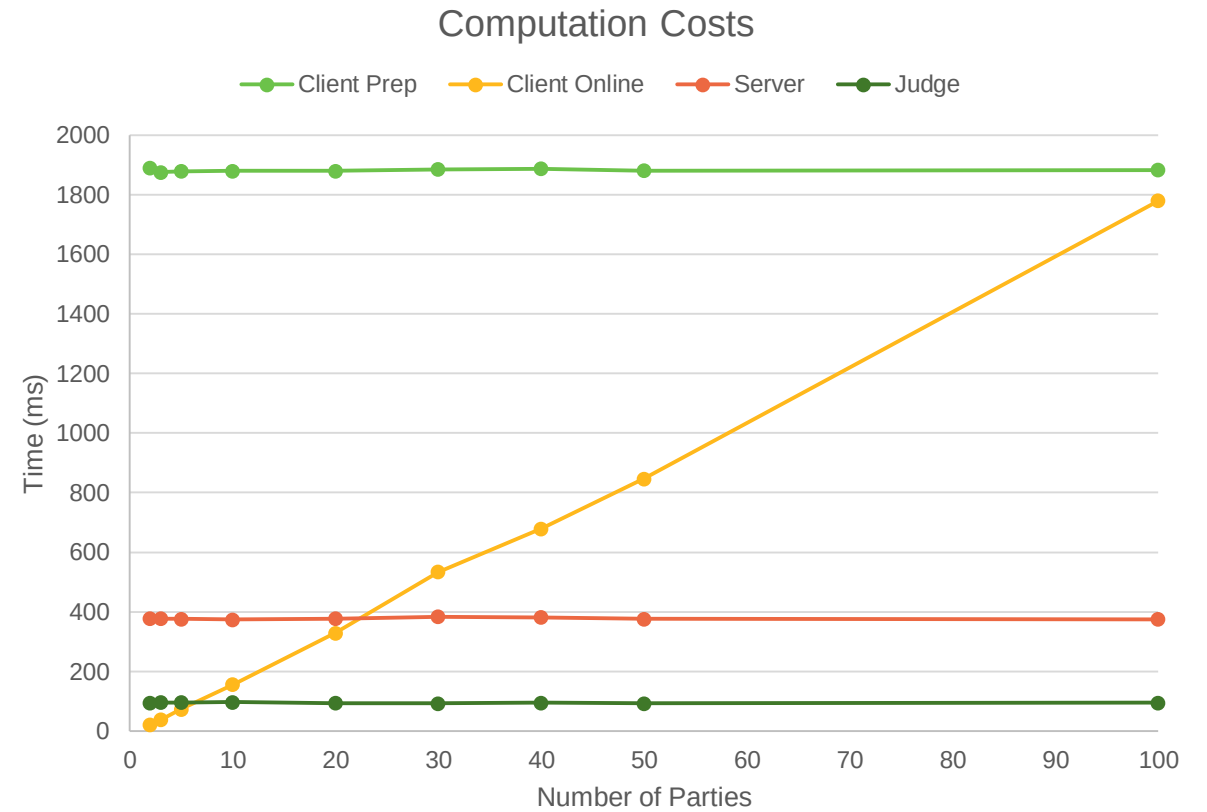
$\xleftarrow{\mathcal{S}_2, GBF_2}$
 $\xleftarrow{\mathcal{S}_i, GBF_i}$
 $\mathcal{S}_{agg} \leftarrow \prod_{i=1}^{t-1} \mathcal{S}_i$
for $v \in [m]$ **do**
 $GBF_{agg}[v] \leftarrow \prod_{i=1}^{t-1} GBF_i[v]$
 $\xleftarrow{\mathcal{S}_{agg}, GBF_{agg}}$

$I = \{\}$
for $j \in [n_r]$ **do**
 $\hat{x}_j \leftarrow e(\sigma_j, \mathcal{S}_{agg})$
 $\hat{y}_j \leftarrow \prod_{u=1}^k GBF_{agg}[h_u(\mathcal{S}_t[j])]$
if $\hat{x}_j = \hat{y}_j$ **do**
 $I = \{\mathcal{S}_t[j]\} \cup I$
 Output I

Baseline

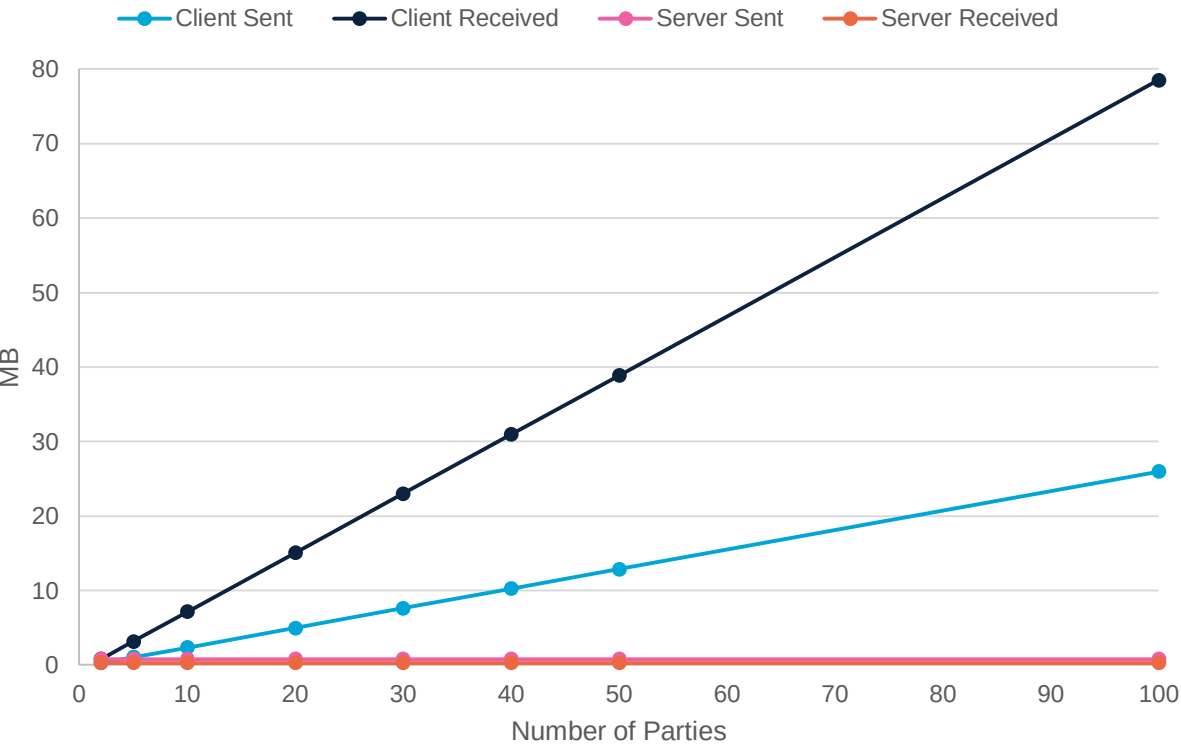


APSI5



Baseline

Communication Costs



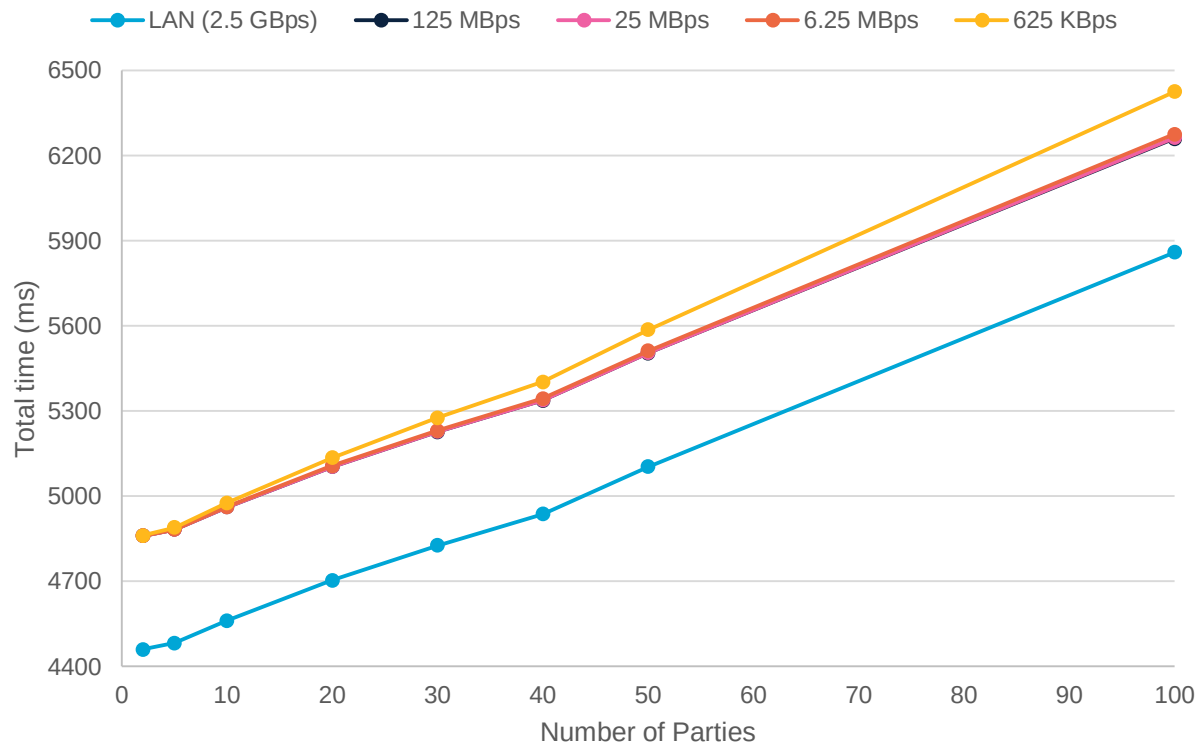
APSI5

Communication Costs



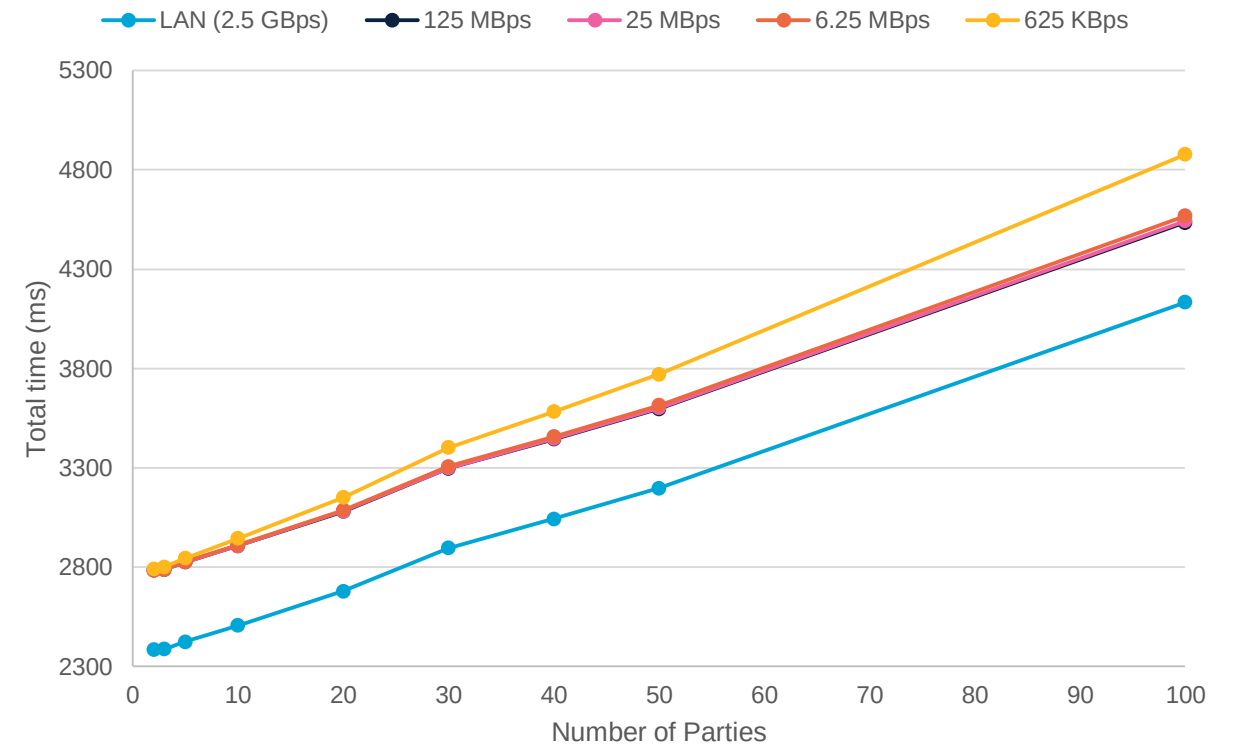
Baseline

Simulated Total Execution Time



APSI5

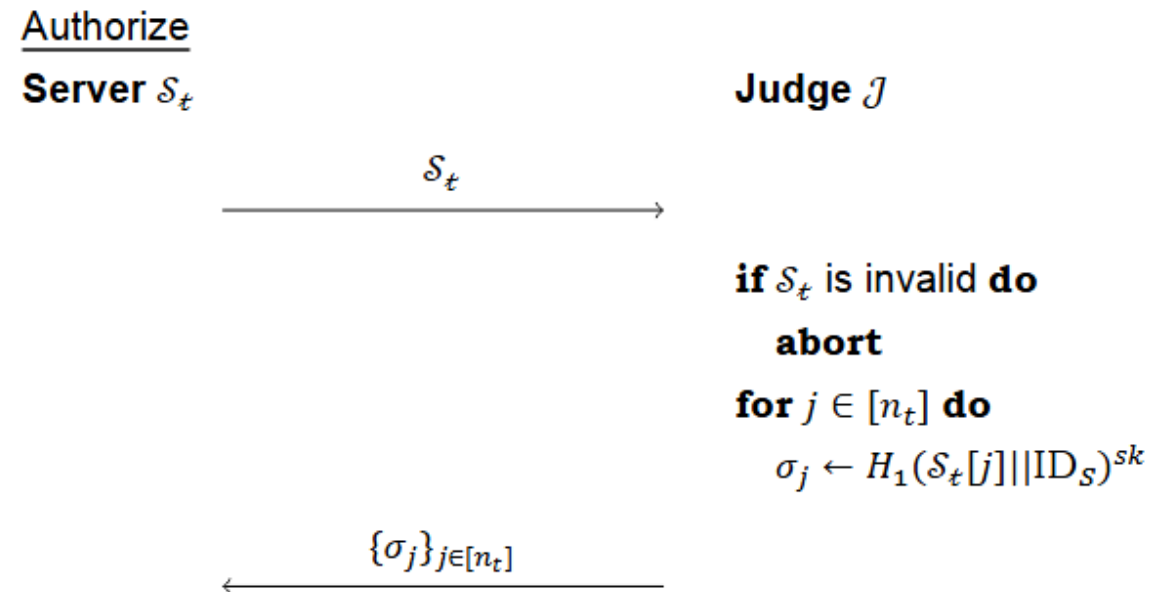
Simulated Total Execution Time



Progress

APSI 4 – Lilika's Judge

- Authorization is identical to [4], except the server will authorize their set since they are the one receiving the intersection



Progress

APSI 4 – Lilika’s Judge

- The server has the secret (decryption) key, so they can decrypt the combined *ERBF*
- Infer *some* information from seeing how many bins are 1

Intersect
Server S_t

Client 1 S_1

$s \leftarrow \mathbb{Z}_q$

...

Client i S_i

s

s

for $j \in [n_1]$ **do**

$\hat{y}_{1,j} \leftarrow e(H_1(S_1[j] || ID_S), pk^S)$

$BF_1 \leftarrow \text{ComputeBF}(\{\hat{y}_{1,j}\}_{j \in [n_1]})$

RBF_1

$ERBF_1$

$ERBF_2$

$ERBF_i$

$S \leftarrow g_2^S$

for $v \in [m]$ **do**

$ERBF_{agg}[v] \leftarrow \prod_{i=1}^{t-1} ERBF_i[v]$

$S, ERBF_{agg}$

$I = \emptyset$

for $j \in [n_t]$ **do**

$\hat{x}_j \leftarrow e(\sigma_j, S)$

$c_j \leftarrow 1$

for $u \in [k]$ **do**

$c_j \leftarrow c_j \times ERBF_{agg}[h_u(\hat{x}_j)]$

if $\text{Dec}(c_j) = 1$ **do**

$I = \{S_t[j]\} \cup I$

Output I

Progress

APSI 4 – Lilika’s Judge

- The server has the secret (decryption) key, so they can decrypt the combined *ERBF*
 - Infer *some* information from seeing how many bins are 1
 - Hence, we can do joint decryption instead (adds one round of interaction) – the server selects the appropriate bins for each element, multiplies them, sends them to the clients, the clients jointly decrypt
 - Change threat model to (augmented) semi-honest server?

Intersect
Server S_t

Client 1 S_1

$s \leftarrow \mathbb{Z}_q$

...

Client i S_i

s

s

for $j \in [n_1]$ **do**

$\hat{y}_{1,j} \leftarrow e(H_1(S_1[j] || ID_S), pk^S)$

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RBF_1

$ERBF_1$

$ERBF_2$

$ERBF_i$

$S \leftarrow g_2^S$

for $v \in [m]$ **do**

$ERBF_{agg}[v] \leftarrow \prod_{i=1}^{t-1} ERBF_i[v]$

$S, ERBF_{agg}$

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for $j \in [n_t]$ **do**

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$c_j \leftarrow 1$

for $u \in [k]$ **do**

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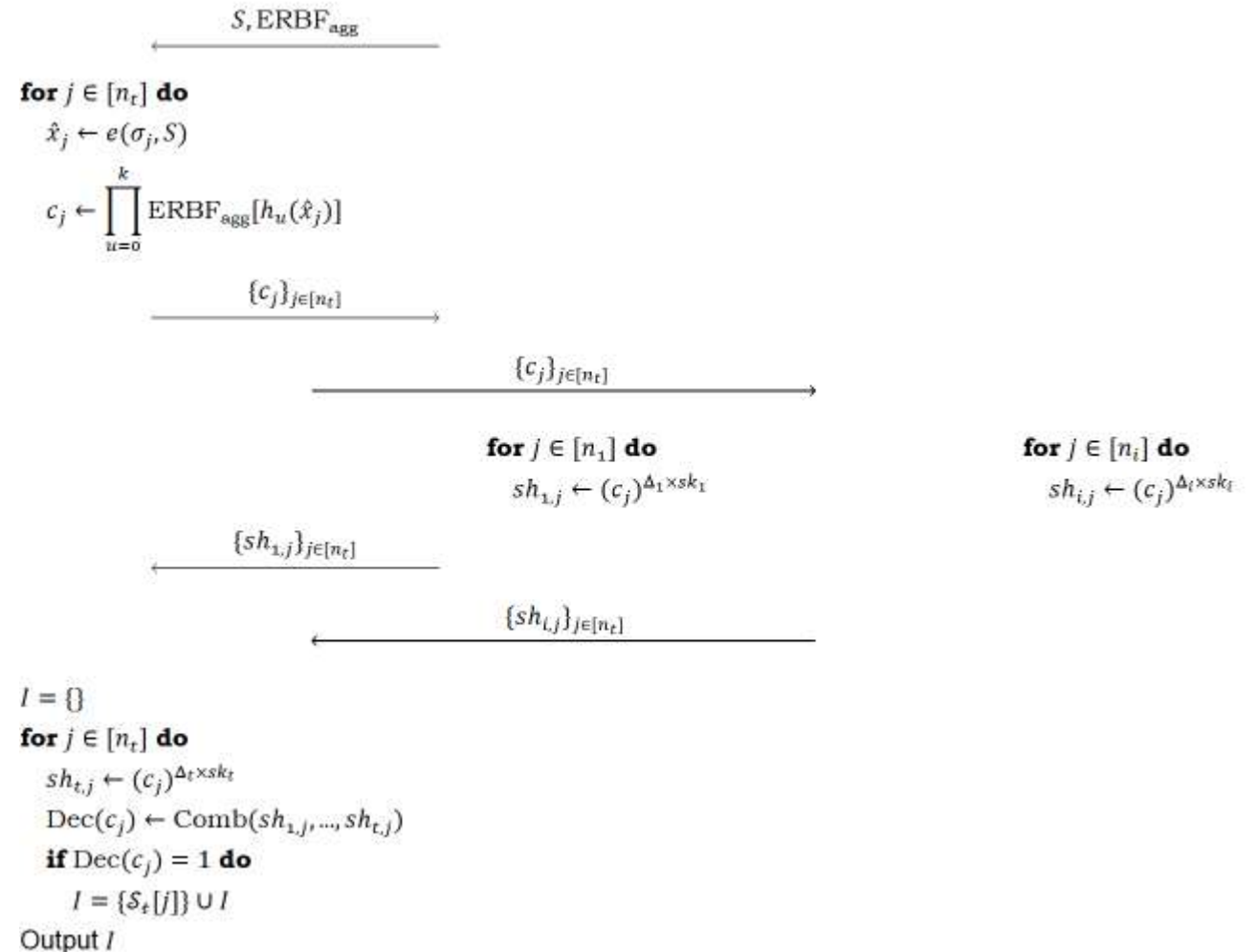
$I = \{S_t[j]\} \cup I$

Output I

Progress

APSI 4 – Lilika’s Judge

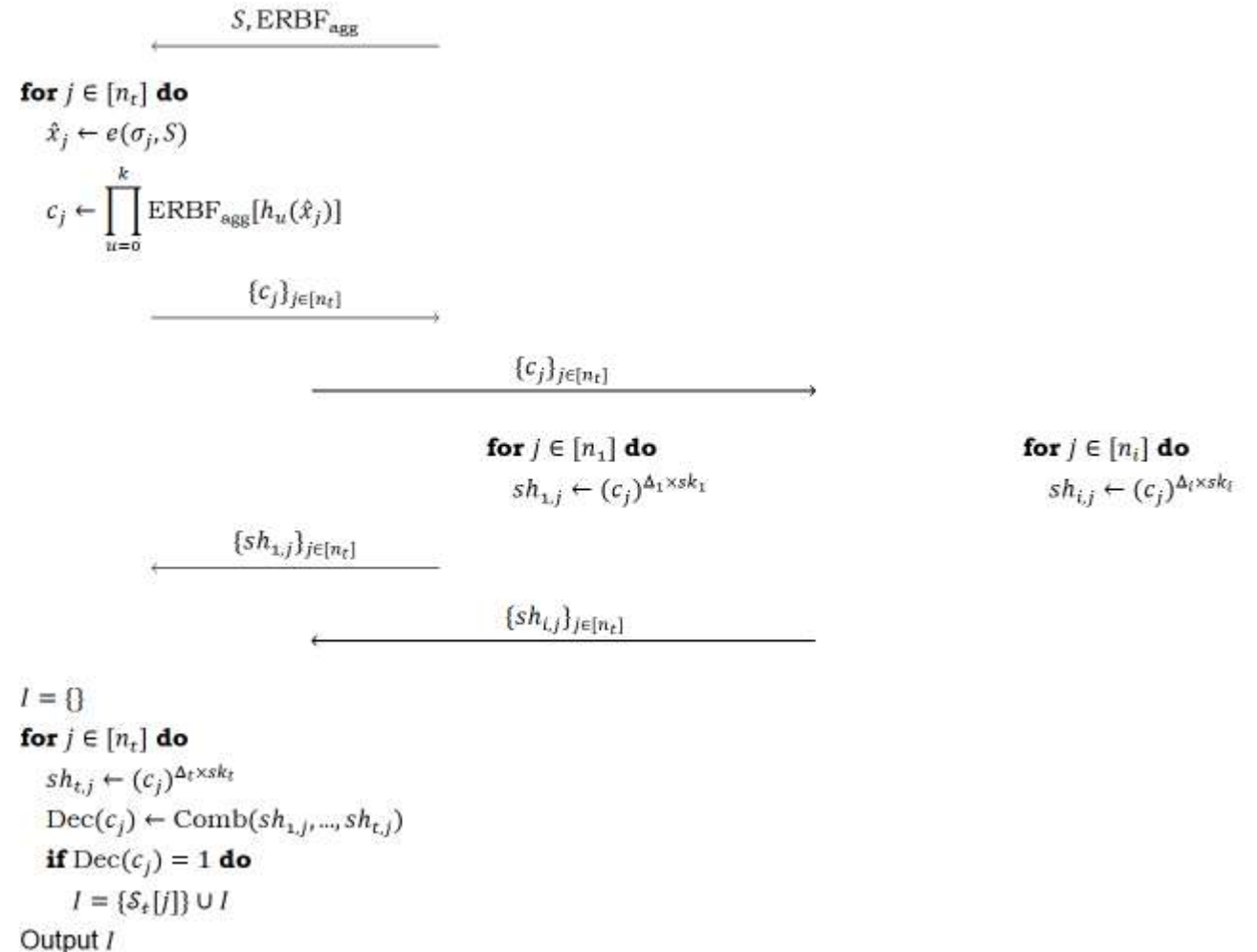
- The server has the secret (decryption) key, so they can decrypt the combined *ERBF*
- Infer *some* information from seeing how many bins are 1
- Hence, we can do **joint decryption instead** (adds one round of interaction) – the server selects the appropriate bins for each element, multiplies them, sends them to the clients, the clients jointly decrypt
- Change threat model to **(augmented) semi-honest server?**



Progress

APSI 4 – Lilika's Judge

- Encrypting the bilinear pairing will take even more space (2 ciphertexts per plaintext with ElGamal) – more communication



Next Week

- Writing!
- Implement APSI4/PAPSI4/PAPSI5

Thank you!

30-04-2026